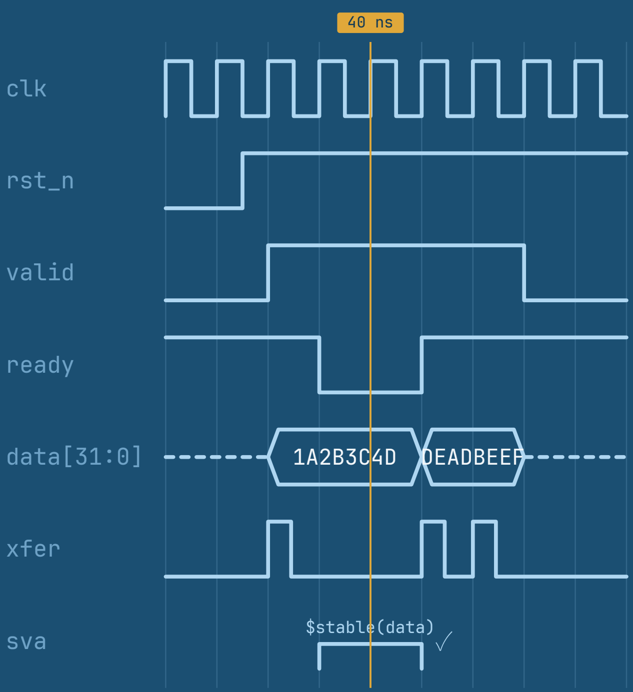
OPEN EDITION · v0.1

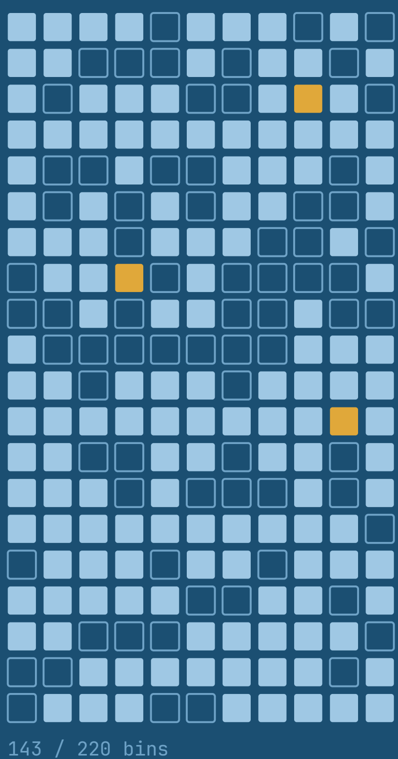
Design Verification

From First Principles to Agentic AI



SVA

```
assert property (@(posedge clk) disable iff (!rst_n)
valid &&!ready |=>$stable(data));
```



Mayur Kubavat

DESIGN VERIFICATION · FROM FIRST PRINCIPLES TO AGENTIC AI

Verification is where silicon meets reality. Every chip that ships has been proven right, or wrong, by an engineer who asked the question the designer did not. This book teaches that discipline from the ground up: how simulators actually schedule events, how to turn a specification into a plan you can measure, how to build SystemVerilog and UVM environments that scale from a block to an SoC, and when to reach for formal, emulation, or portable stimulus instead. It then looks forward, to Python and SystemC flows, to the open-source toolchain, and to the machine learning and agentic AI that are reshaping how verification gets done. Every listing is runnable, every chapter ends with exercises, and the whole book is free to read.

INSIDE THIS BOOK

- Foundations: planning, coverage models and simulator internals
- SystemVerilog, assertions and UVM, from first principles to SoC-scale reuse
- Formal, emulation, low-power, CDC and portable stimulus
- cocotb, pyuvm, SystemC and the open-source toolchain
- Machine learning and agentic AI applied to real verification work

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Mayur Kubavat is a design verification engineer whose work spans high-speed interconnect verification, including PCIe 6.0 verification IP, and open-source verification tooling.

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Source, examples and errata: github.com/mayurkubavat/dv-handbook

web edition